```
###### QUESTION 1############

# Load MASS package
library(MASS)
data("crabs")
str(crabs)
table(crabs$sp)
table(crabs$sex)
table(crabs$sp, crabs$sex)
barplot(table(crabs$sp, crabs$sex),
        beside = TRUE,
        legend = TRUE,
        main = "Bar Chart of Species and Sex",
        xlab = "Species",
        ylab = "Frequency")

mosaicplot(table(crabs$sp, crabs$sex),
           main = "Mosaic Plot of Species and Sex",
           color = TRUE)

# Load MASS package and dataset
library(MASS)
data(crabs)


# Create a grouping variable combining species and sex
group <- interaction(crabs$sp, crabs$sex)

# Side-by-side boxplots of body depth (BD) for the four groups
boxplot(crabs$BD ~ group,
        main = "Side-by-Side Boxplots of Body Depth by Species and Sex",
        xlab = "Species and Sex Groups",
        ylab = "Body Depth (BD)",
        col = c("lightblue", "lightpink", "lightgreen", "lightyellow"))

by(crabs$BD, group, boxplot.stats)

aggregate(crabs$BD, by = list(Group = group), FUN = mean)
aggregate(crabs$BD, by = list(Group = group), FUN = median)

aggregate(crabs$BD, by = list(Group = group), FUN = sd)
aggregate(crabs$BD, by = list(Group = group), FUN = IQR)


### QUESTION 3#####################################
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# Load MASS package and dataset
library(MASS)
data(crabs)

# Histogram of carapace length (CL) for each species side by side
par(mfrow = c(1, 2))

# Histogram for species B
hist(crabs$CL[crabs$sp == "B"],
     main = "Histogram of Carapace Length (Species B)",
     xlab = "Carapace Length (CL)",
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    col = "lightblue",
    breaks = 10)

# Histogram for species O
hist(crabs$CL[crabs$sp == "O"],
    main = "Histogram of Carapace Length (Species O)",
    xlab = "Carapace Length (CL)",
    col = "lightgreen",
    breaks = 10)

par(mfrow = c(1, 1))

# Optional: Check skewness numerically (requires moments package)
# install.packages("moments")
library(moments)
skewness(crabs$CL[crabs$sp == "B"])
skewness(crabs$CL[crabs$sp == "O"])

### QUESTION 4######################################
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# Load MASS package and dataset
library(MASS)
data(crabs)

means <- aggregate(cbind(FL, RW, CL, BD) ~ sp + sex, data = crabs, FUN = mean)
print(means)
mean_matrix <- as.matrix(means[, 3:6])

group_labels <- paste(means$sp, means$sex, sep = "-")
mean_matrix_t <- t(mean_matrix)

# Grouped bar chart
barplot(mean_matrix_t,
        beside = TRUE,
        legend = rownames(mean_matrix_t),
        names.arg = group_labels,
        main = "Grouped Bar Chart of Mean Morphological Measurements",
        xlab = "Species and Sex Groups",
        ylab = "Mean Measurement",
        col = c("lightblue", "lightgreen", "lightpink", "lightyellow"))

# Compute species means only
species_means <- aggregate(cbind(FL, RW, CL, BD) ~ sp, data = crabs, FUN = mean)
print(species_means)

# Compute sex means only
sex_means <- aggregate(cbind(FL, RW, CL, BD) ~ sex, data = crabs, FUN = mean)
print(sex_means)

# Calculate species differences
species_diff <- abs(species_means[1, 2:5] - species_means[2, 2:5])
print(species_diff)

# Calculate sex differences
sex_diff <- abs(sex_means[1, 2:5] - sex_means[2, 2:5])
print(sex_diff)
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names(which.max(species_diff))
names(which.min(sex_diff))


### QUESTION 5 ###########################################
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# Load MASS package and dataset
library(MASS)
data(crabs)
measurements <- crabs[, c("FL", "RW", "CL", "CW", "BD")]

# Produce scatterplot matrix (pairs plot)
pairs(measurements,
    main = "Scatterplot Matrix of Crab Measurements",
    pch = 19,
    col = "blue")

pairs(measurements,
    main = "Scatterplot Matrix of Crab Measurements by Species",
    pch = 19,
    col = as.numeric(crabs$sp))

cor(measurements)


### QUESTION 6#################################################
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